

Aayush Tyagi

Software Engineer | Java | Full Stack (MERN)

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SUMMARY

Full-Stack Developer with hands-on experience in **Java**, **DSA**, and **MERN stack** development. Proficient in building **AI-integrated web platforms**, scalable **REST API** design, and **JWT-based authentication**. **AWS Certified Cloud Practitioner** with 350+ problems solved on LeetCode and multiple production applications deployed with optimized performance. Seeking software engineering roles to build high-impact, scalable technology solutions.

EDUCATION

Meerut Institute of Engineering and Technology <i>B.Tech in Computer Science and Engineering</i> GPA: 9.0/ 10.0 (upto 6th Semester)	2023 – 2027 Meerut, India
Guru Govind Singh Public School <i>Class 12th</i> — Percentage: 92%	2022 – 2023
Guru Govind Singh Public School <i>Class 10th</i> — Percentage: 81%	2020 – 2021

SKILLS

Languages: Java | C & Python(Basic) | SQL | JavaScript

Frontend: React.js | Tailwind CSS | HTML5 | CSS3 | Bootstrap

Backend: Node.js | Express.js | REST APIs

Databases: MongoDB | MySQL

Core CS: OOPs | DBMS | Computer Networks | Data Structures & Algorithms | Operating System

Tools: Git | GitHub | Postman | VS Code | AWS | Azure

Soft Skills: Leadership | Collaboration | Creativity | Project Management

EXPERIENCE

Trident Guard – Top 10 Finish, Tally CodeBrewers 2026 Hackathon Jul 2026
React, Node.js, Express.js, WebCrypto API, face-api.js, Authentication [Live Demo](#) | [GitHub](#)

- Built an **enterprise zero-trust authentication platform** preventing account takeovers and insider threats, finishing in the **top 10 teams**.
- Engineered **hardware-bound authentication** using WebCrypto (P-256 ECDSA) and **face-api.js facial recognition** with a 3-strike fallback cascade (Face ID → PIN → TOTP).
- Designed a **zero-storage biometric model**, keeping facial descriptors and private keys in client-side IndexedDB only, never on servers.
- Implemented multi-party authorization via **2-of-3 Shamir Social Recovery** and a Duress Canary PIN that silently alerts security under coercion.

Full Stack Web Developer Intern (Eduskills – Remote) Jan 2026 – Apr 2026
Frontend, Backend, Database Technologies, RESTful APIs

- Completed comprehensive training in Full Stack Web Development covering modern **frontend, backend, and database** technologies.
- Built responsive web applications and engineered robust **RESTful APIs** using contemporary development frameworks.
- Implemented scalable **end-to-end application solutions** while managing complex database integrations.
- Developed a practical understanding of the **software development lifecycle (SDLC)** and automated deployment workflows.

PROJECTS

HiveMind (Multi-Agent AI Platform System) 2026
MERN, LangGraph, Groq, DeepSeek, Redis, Qdrant, Render, Vercel [Live Demo](#)

- Architected a **Multi-Agent AI Engine with LangGraph**: Built an autonomous state machine routing queries to specialized agents (Code, Search, PPT, Vision) using **Groq** and **DeepSeek**, cutting response latency by **~65%** with dynamic multi-provider failover.
- Engineered a **5-Microservice Distributed Backend**: Decoupled Gateway, Auth, Agent, Chat, and Billing services on **Render**; integrated **Upstash Redis** for atomic rate limiting and sliding-window memory caching to reduce DB read load by **80%**.
- Implemented **PDF RAG & Production Cloud Deployment**: Vectorized documents into **Qdrant Vector DB** for semantic similarity search; configured cross-origin **SameSite=None** auth and deployed live across **Vercel** and **Render**.

SymptoGenie (AI Healthcare Assistant App) Jan 2026 – Apr 2026
Node.js, Express, React.js, Overpass API [Live Demo](#)

- Engineered an **AI-powered symptom analysis engine** using weighted scoring and medical ontologies for common ailment triage.
- Built an **OCR/NLP pipeline** to extract and structure data from medical reports (PDF/Images) using Tesseract and spaCy.
- Developed scalable **Node.js/Express REST APIs** with modular routing, input validation, and centralized error handling.
- Integrated interactive geolocation maps using **React-Leaflet** and **OpenStreetMap (OSM)** tiles with real-time hospital discovery powered by the **Overpass API**.
- Implemented **JWT authentication** and optimized MongoDB schemas with proper indexing to reduce query overhead.

MelodyVerse (AI Powered Music Streaming App)
Node.js, Express, React.js, Google Gemini API

Mar 2026 – May 2026
[Live Demo](#)

- Developed “MelodyBot,” a **voice assistant** powered by Google Gemini 2.5 Flash for natural-language music queries.
- Integrated **face-api.js** for real-time facial emotion detection, categorizing **8 core moods** to curate playlists.
- Engineered a server-side caching layer with Redis to minimize redundant API calls and reduce response latency.
- Built a **MERN stack streaming platform** with Context API for persistent playback with zero audio interruption.
- Designed a responsive, premium UI with real-time search indexing, achieving a **Lighthouse performance score of 95+**.

TaskFlow (Task Management Web Application)
HTML, CSS, JavaScript

Jan 2025 – Feb 2025
[Live Demo](#)

- Developed a comprehensive task management web application to organize, prioritize, and track daily activities efficiently.
- Enabled persistent data storage and dynamic, real-time task updates using JavaScript and browser **LocalStorage**.
- Designed a highly responsive and intuitive user interface focused entirely on enhancing productivity and usability.

CERTIFICATIONS

- [AWS Certified Cloud Practitioner](#)
- [Microsoft Certified \[AZ-900\]: Azure Fundamentals](#)
- [AWS Educate: Introduction to Generative AI](#)
- [Oracle Certified AI Foundations Associate](#)

ACHIEVEMENTS

- Placed in the **Top 10 teams** at **Tally CodeBrewers 2026 Hackathon** for building **Trident Guard**, an enterprise zero-trust authentication platform with hardware-bound biometric security.
- Qualified **Round 1** and **Round 2** of **Flipkart Grid 8.0 2026**, a national-level tech competition.
- Solved **350+ problems** on LeetCode, strengthening problem-solving and algorithmic skills.
- Deployed **3 production-grade full-stack applications** with live demos, end-to-end authentication, and database integration.
- Active **member of GDG MIET**, contributing to technical events, workshops, and developer community initiatives.